

Generative AI Project Paper Guidelines

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Abstract. The title should clearly reflect the problem being addressed and the approach used. The abstract must be written as one compact paragraph that briefly introduces the problem, describes the method or models implemented, highlights the key results obtained, and states the main findings or conclusions of the study. The abstract should briefly summarize the contents of the paper in 150–250 words.

Keywords: Generative AI · Voice AI · Diffusion Models · Transformers.

1 Introduction

The introduction should be written in two to three well-structured paragraphs. It should begin with a general description of the problem domain, explaining the motivation and importance of the research. This should be followed by a discussion of the research gap or limitation in existing approaches. Finally, the main contributions of the paper should be clearly stated in two to three concise bullet points within the paragraph to highlight what has been achieved in this work.

2 Literature Review

The literature review should include a discussion of eight to ten relevant and recent research papers. It should be written in four to five paragraphs, where each paragraph synthesizes related work rather than listing papers individually. Students should describe the contributions of prior work, identify their limitations, and clearly position their own work in relation to these studies.

3 Materials and Methods

3.1 Dataset and Dataset Preparation

This subsection should describe the dataset or datasets used in the study. Students should explain the source of the data, provide examples of the data (such as images or text samples where applicable), and describe all preprocessing steps applied. It should also include a discussion of the data distribution and a table summarizing the total number of samples, as well as the split between training, validation, and testing sets.

3.2 Methodology

This subsection should describe the models used in the study. Students should explain the architecture and working mechanism of each model, along with relevant mathematical formulations and key equations. The training procedure should also be described, including loss functions, optimization techniques, and important hyperparameters. If a novel method is proposed, its contribution should be clearly highlighted and differentiated from existing approaches.

4 Experimental Results

4.1 Experimental Setup

This subsection should describe the complete experimental environment. Students should include details about the hardware used (such as CPU, GPU, and RAM specifications), the software frameworks and libraries used for implementation, and the execution environment (such as Google Colab or local systems). The description should be sufficient to ensure reproducibility of the experiments.

4.2 Evaluation Metrics

This subsection should describe the evaluation metrics used to assess model performance. Students should explain each metric in detail, justify why it is appropriate for the problem, and include mathematical definitions or equations where applicable.

4.3 Results and Analysis

This subsection should present all experimental results in a structured and comprehensive manner. Students must include plots such as training versus validation curves against epochs, error plots, and any other relevant visualizations. Quantitative results should be presented in tables that allow direct comparison between models. Visual outputs, such as generated samples or predictions, should also be included where applicable. In addition to presenting results, students must provide a detailed analysis explaining the observed trends, comparing model performance, discussing strengths and limitations, and highlighting key insights derived from the experiments.

5 Conclusion

The conclusion should be written as a single paragraph summarizing the key findings of the study. It should highlight the main contributions, identify the best-performing model or approach, and briefly suggest possible directions for future work.

6 References

All references must follow the Springer format and be complete with all necessary details. Students should maintain a separate BibTeX (.bib) file and use it within Overleaf for proper citation management. Incomplete or incorrect references should be strictly avoided.